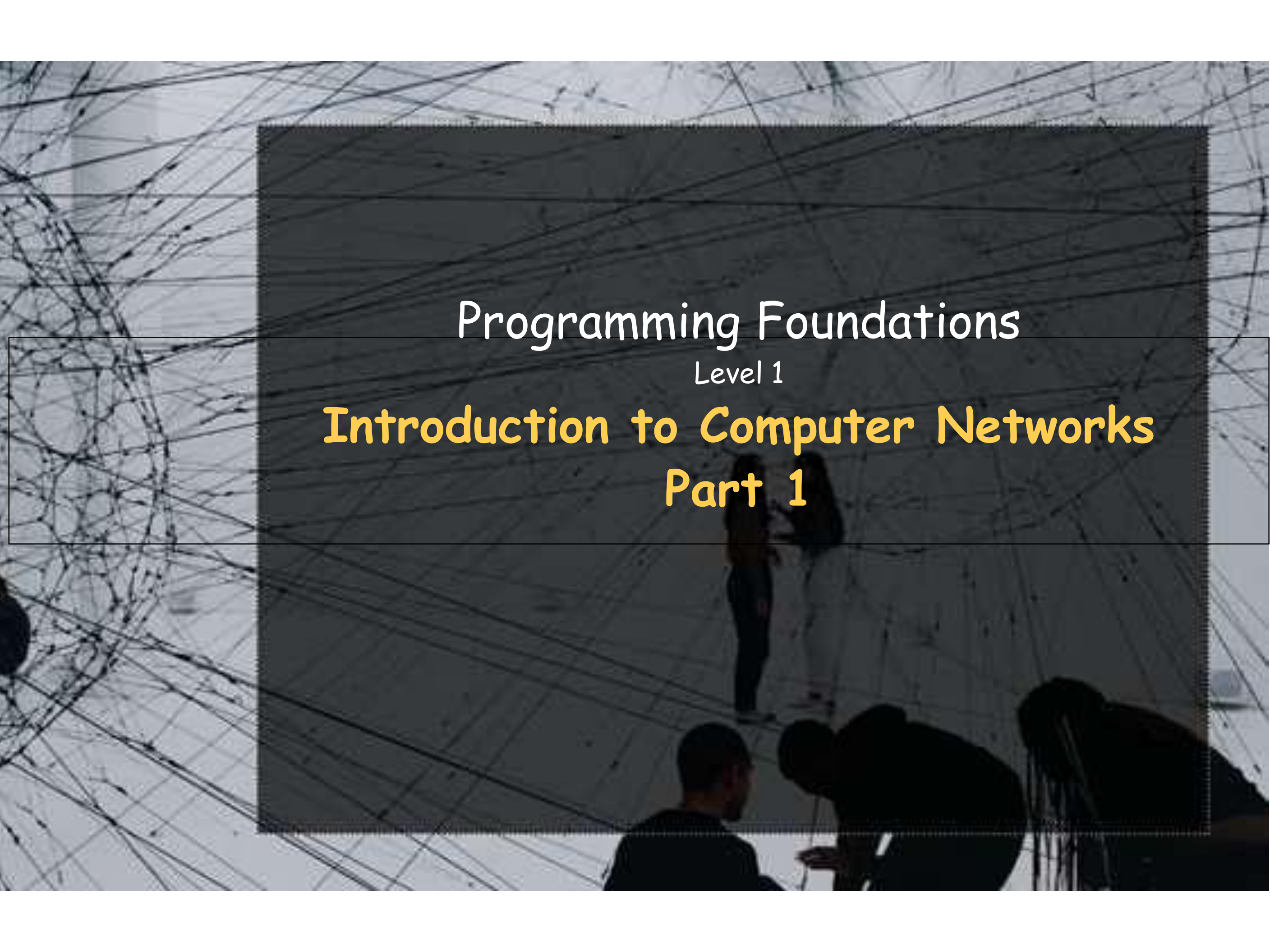




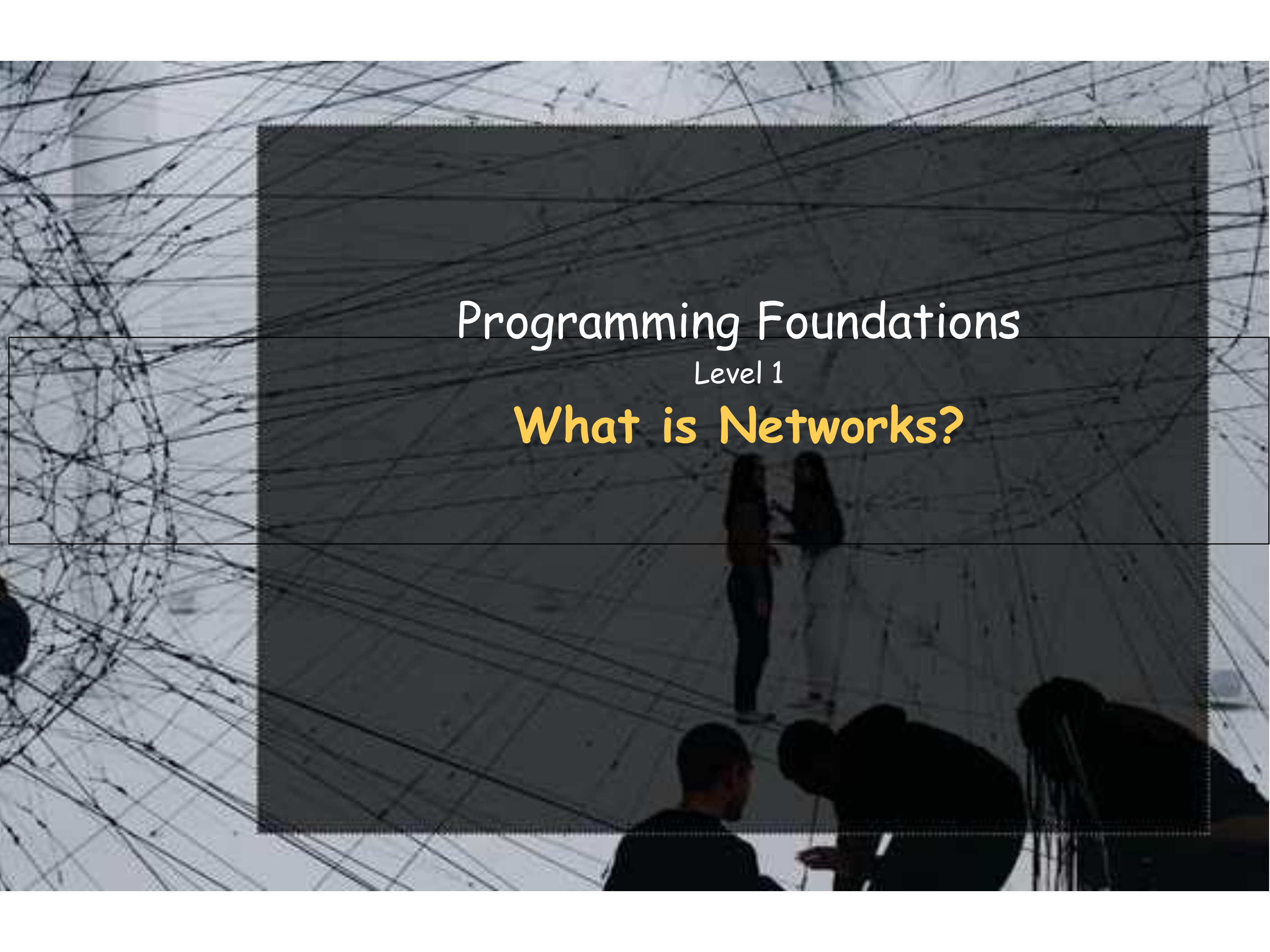
Programming Foundations Level 1



Programming Foundations

Level 1

Introduction to Computer Networks Part 1



Programming Foundations

Level 1

What is Networks?

Networks Part I — Basics, Types, and How Devices Communicate

A network is simply a group of connected computers and devices that can communicate and share resources. From your home Wi-Fi to the global internet, networks enable messaging, streaming, printing, remote control, and more. In this lesson you'll learn why we need networks, the difference between LAN and WAN, and how devices communicate via Ethernet, Wireless, Protocols, and Wi-Fi.

WHAT IS A NETWORK?

NETWORK:

a group of connected computers and devices.



STAY CONNECTED



WHY DO WE NEED NETWORKS?

Share files, printers, and internet access.



Communicate (messages, calls streaming).



Communicate (mevices, streaming).



Control smart devices (TV, speakers, of, lights) from your phone.



Collaborate at school, work, or across locations.



STAY CONNECTED



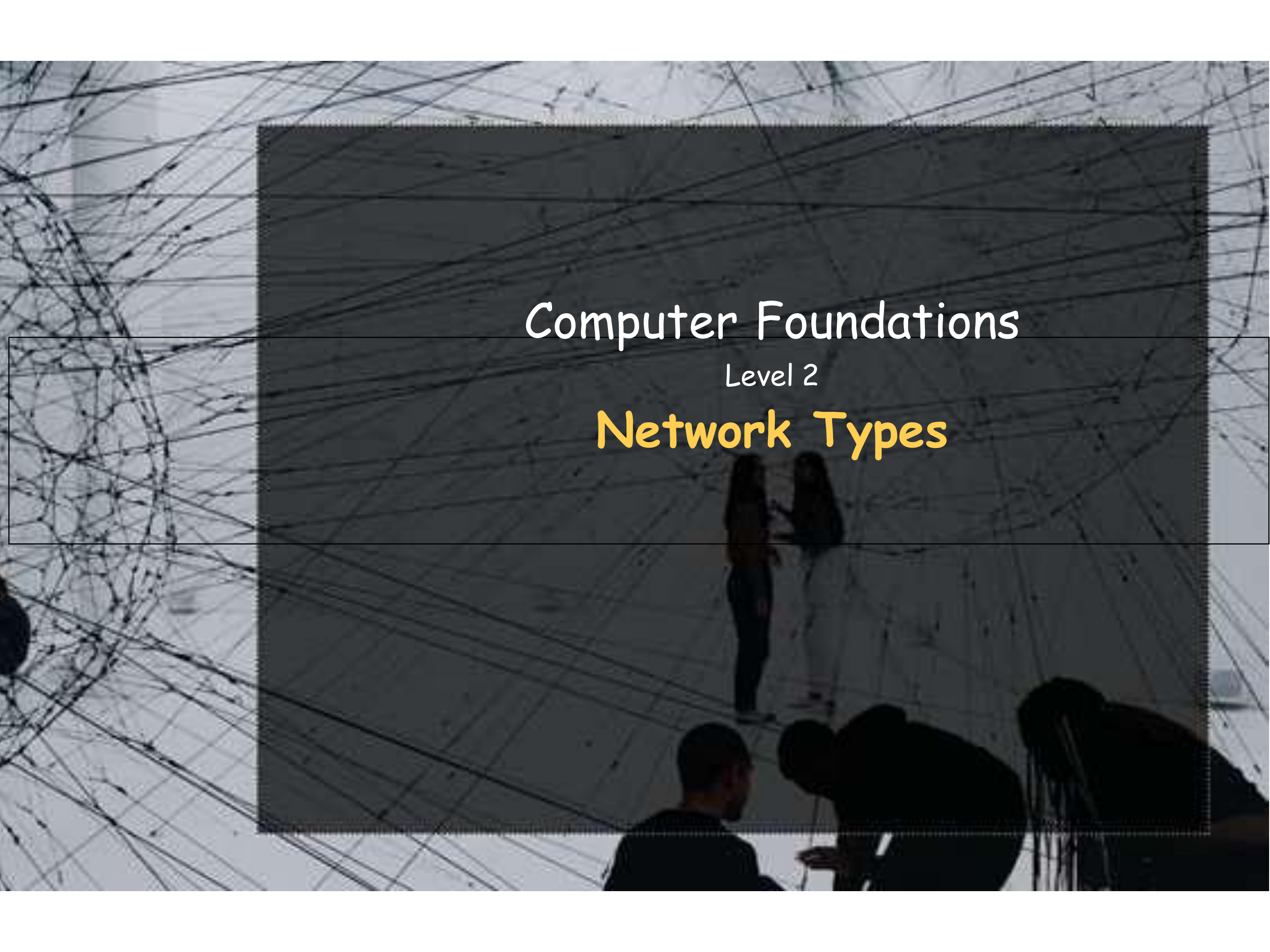
What is Network?

Network: is a group of connected computer and devices.

- For example, the computers, phones, and other devices that your family owns are all part of your home computer network.
- When you get a new tablet and enter in the Wi-Fi password, you're joining that tablet to your home computer network.

Why do we need Network?

- Devices in the same network can easily communicate and share resources with each other.
- For example : you can control your TV with your mobile, or you can stream videos from your mobile to your TV.



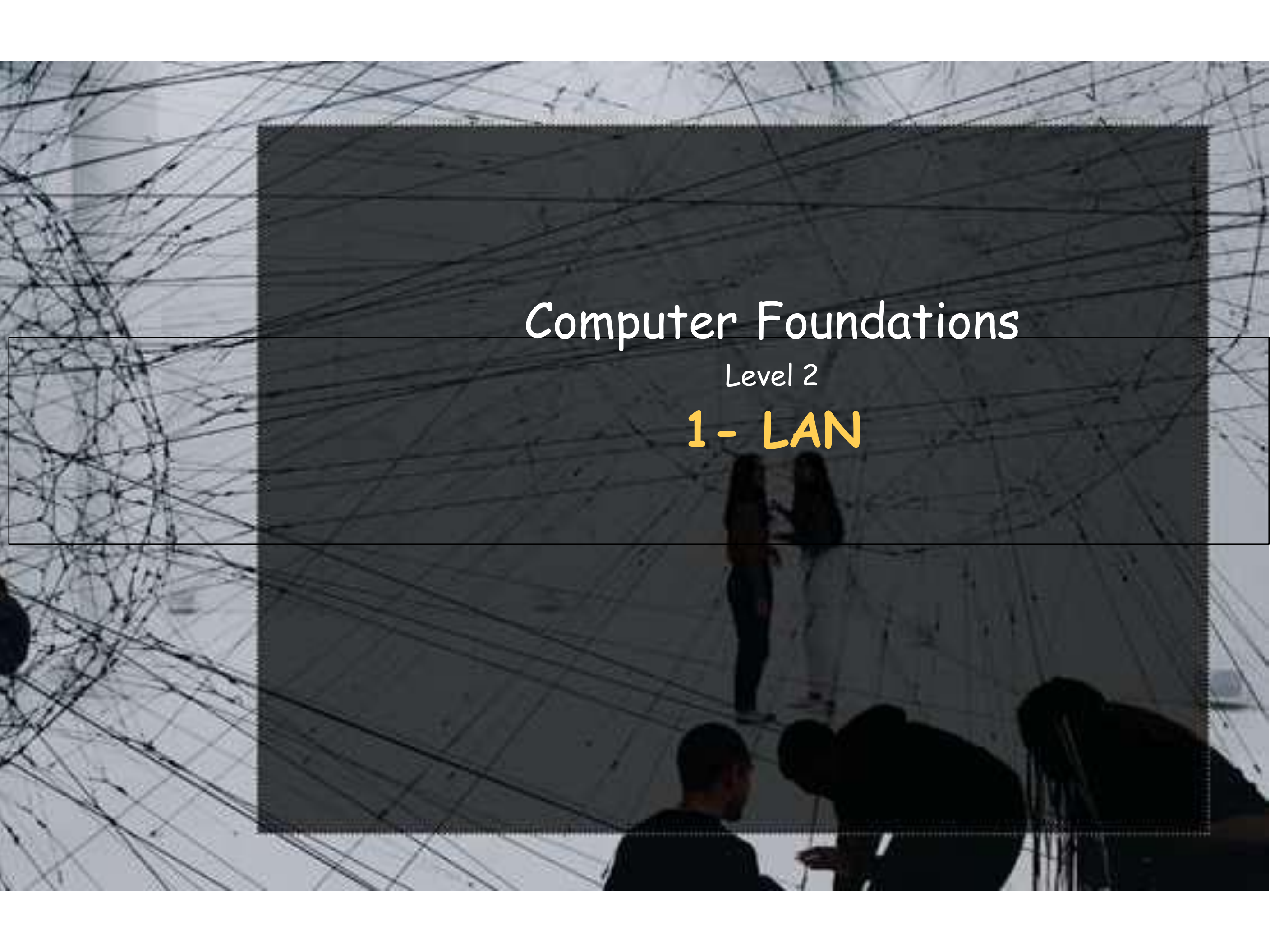
Computer Foundations

Level 2

Network Types

Network Types





Computer Foundations

Level 2

1- LAN

What is LAN?

LAN: Local Area Network that has a group of connected devices.

- LAN is a type of network made up of nearby computers.
- A LAN can have anywhere from a few devices (like a home network) to hundreds of devices (like a school or hospital)
- Large organizations like colleges can have multiple LANs - maybe one per building.
- A **nearby group** of connected devices (home, classroom, office, hospital).
- Can range from a **few** devices (home) to **hundreds** (campus building).
- Big organizations can have **multiple LANs** (e.g., one per building).



Computer Foundations

Level 2

Communicating on LAN

◆ How Do Computers Communicate on a LAN?

Communication happens over **wired** or **wireless** connections, following agreed rules called **protocols**:

- **Ethernet (wired)**: rules for sending data **through cables** (e.g., via a **switch** or **router**).
- **Wireless**: rules for sending data **through radio signals**; common form is **Wi-Fi**.
- **Protocol**: a **standard set of rules** defining how devices **format, send, and receive** data so everything understands each other. the **language/rules** devices follow to communicate.
- **Wi-Fi (Wireless Fidelity)**: a **popular wireless method** to connect to a network. **radio-based LAN rules**.
- **Router**: connects your LAN to the **outside** (internet/other networks).
- **Switch**: interconnects devices **inside the LAN**.

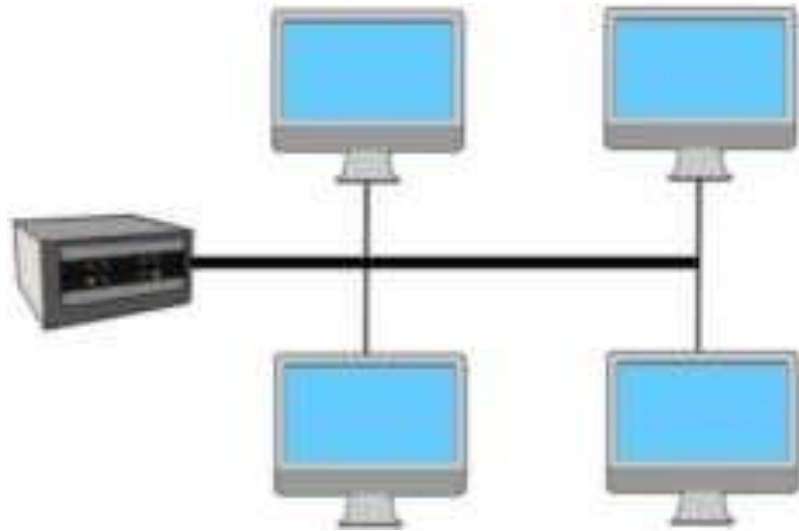
In practice, a **router** connects your LAN to other networks (and the internet), while a **switch** efficiently interconnects devices inside your LAN.

Connection Types



Communication Ways

Switch



Ethernet

Rules that computers follow for communicating on a wired network



Router

Wireless

Rules that computers follow for communicating through a radio signals

Wi-Fi is a popular way to connect wirelessly to a network.

Wi-Fi: Wireless Fidelity

Those Rules are called:

Protocol

Protocol: is a standard set of rules for how devices communicate through wires or wireless



Computer Foundations

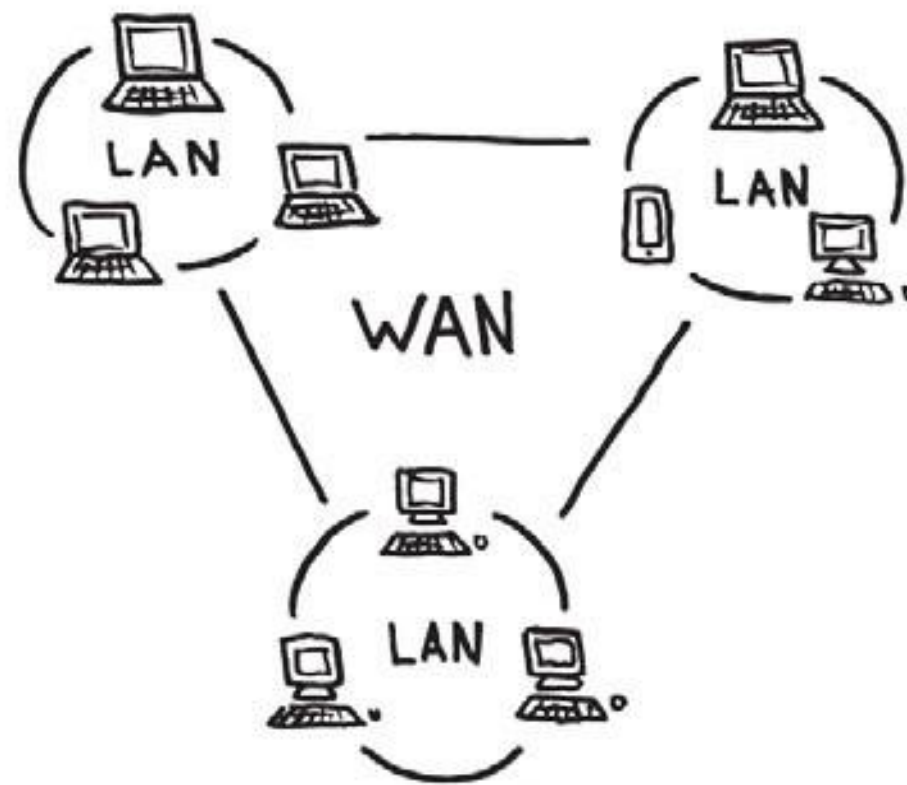
Level 2

2 - WAN

What is WAN?

WAN: Wide Area Network.

- WAN is made of connected LANs that are far from each other (like in different parts of the country or world).
- WANs connect thousands or even millions of devices.
- Many LANs connected across cities, countries, or the world.
- WANs can connect thousands or millions of devices.
- The internet is the biggest WAN.



Interconnection

- ◆ **Network = connected devices** that communicate and **share resources**.
- ◆ **LAN (local) vs WAN (wide)** → **scope & distance** differ; WANs interconnect **many LANs**.
- ◆ **Ethernet (wired)** and **Wi-Fi (wireless)** use **protocols (rules)** so devices **understand** each other.

By understanding these links, learners can map home and school setups to the larger internet and reason about how data flows end-to-end. 🚀

Activity

Instructions:

- 1 What is a **Network**?
- 2 Why do we **need** a **Network**?
- 3 What is the difference between **LAN** and **WAN**?
- 4 How do **computers** **communicate** together on a network?
- 5 What is **Ethernet**?
- 6 What is **Wireless**?
- 7 What is a **Protocol**?
- 8 What is **Wi-Fi**?
- 9 **Wi-Fi** stands for what?



Activity Solution



Solution:

1 Network:

A group of **connected computers and devices** that can communicate and share resources such as files, printers, or internet access.

2 Why we need a Network:

To **share data, resources, and communicate** efficiently between computers, phones, and other devices.

3 LAN vs WAN:

LAN (Local Area Network): covers a **small area** (home, office, school).

WAN (Wide Area Network): connects **multiple LANs** across cities or countries
the **Internet** is the largest WAN.

4 How computers communicate:

They exchange data through **wired (Ethernet)** or **wireless (Wi-Fi)** connections using shared **communication protocols**.

5 Ethernet:

A **wired networking standard** that defines how computers send and receive data using cables.

6 Wireless:

A **network connection without cables**, using **radio waves** to send and receive data.

7 Protocol:

A **set of rules** that defines how computers communicate — it ensures all devices understand each other when exchanging data.

8 Wi-Fi:

A **wireless networking technology** that allows devices to connect to a network and the internet **without cables**.

9 Wi-Fi stands for:

Wireless Fidelity.

Conclusion

- ✓ A **network** connects devices so they can **communicate** and **share resources**.
- ✓ **LAN** covers **local** spaces; **WAN** connects **many LANs** across long distances.
- ✓ Devices communicate via **Ethernet (wired)** or **Wireless (Wi-Fi)** under agreed **protocols**.
- ✓ Knowing these basics lets you understand home setups, campus networks, and the broader internet.

Thank you 😊